

Spectral Non-Concentration Implies Global Regularity for 3D Navier-Stokes on T^3

Jonathan Simons. This record (10.5281/zenodo.20272545) is a 2026 exploratory draft. It remains published and citable.¹ This page is the public-facing face of the record. It is not a retraction notice. The original draft follows the errata.

Live exploratory stack (cite these, not prize packaging)

Live exploratory stack: Φ -renormalization 10.5281/zenodo.22050974 (not a Clay Navier-Stokes proof); Route C 10.5281/zenodo.22050963 (not a proof of RH); Ring lemma 10.5281/zenodo.22050976 (Goldbach remains conditional on SND); T2 under SND 10.5281/zenodo.22050965 and Bridge* stay named derived objects; inverse-GCD note 10.5281/zenodo.22050962 is a restricted Rayleigh bound, not a full-spectrum floor.

¹ August 2026 prize-claim language walked back; see errata below / page 2 of this file and status note 10.5281/zenodo.22050978.

Errata: prize-claim language walked back

The pages after this one are the original 2026 draft, unchanged. They are on the record so the walk-back is auditable. The following prize-claim language in that draft does not stand.

Why the August 2026 action happened

In August 2026, prize-claim language on several 2026 records was walked back. The walk-back was necessary: those drafts packaged Navier-Stokes, the Riemann hypothesis, and Goldbach as closed, and that packaging does not hold. The walk-back was executed badly. Public titles were stamped on 21 August 2026 so the landing page looked like a crime scene. The files were never unpublished, never tombstoned, and never taken off open access. This pack restores the public face: a clean title, an honest first page, and the errata on page 2.

Prize-claim language that does not stand

Clay Statement B / Navier-Stokes millennium packaging. The June drafts treated a regularity claim as closed. That language is walked back. The Φ -renormalization note is the live fluids object.

Quantum Lens / Q6 Goldbach as a theorem. Goldbach stays conditional on SND. Q6 is not a proof.

Montgomery-Dyson / pair correlation as a finished RH argument. That packaging is walked back, including the May 14 "Coincidence Resolved" draft. Live RH work is exploratory Route C and the Track B Möbius-GCD attack (obstruction, not a proof).

Three-in-one / $\text{SND} \equiv \text{GNC} \equiv \text{Bridge}$ as closures. Those identities were used as if they closed NS, RH, and Goldbach. They do not. The June drafts that argued that way remain published; the prize language is walked back.

Full-spectrum $\lambda_{\min}(Q_N) > -1/2$ as a RH proof. That bound is not a proof of RH.
Inverse-GCD $1/\text{gcd}$ is not the live Route C operator.

This page is the errata. It is not a tombstone. No Millennium problem is claimed. Domain Architect is a local FRA classifier (inquiry). ChatVault is search. Neither wrote these Zenodo titles.

SPECTRAL NON-CONCENTRATION IMPLIES GLOBAL REGULARITY FOR 3D NAVIER–STOKES ON $\mathbb{T}^3$

JONATHAN R. SIMONS

ABSTRACT. We establish a spectral framework for global regularity of the three-dimensional incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$. The central object is the shell-helical GCD interaction operator $H_N[u(t)]$, whose minimum eigenvalue governs spectral stability of the Leray–Hopf flow.

Our main result (Theorem 4.1) states: *if the Spectral Non-Concentration (SND) condition holds quantitatively — meaning the evolving operator stays within δ_0 of a frozen reference in operator norm — then $\inf_{t \geq 0} \lambda_{\min}(H_N[u(t)]) > -\frac{1}{2}$, precluding spectral blowup for all time.*

The proof reduces to a single application of Weyl’s perturbation inequality once the frozen gap (Theorem 2.1) and operator continuity (Lemma 3.1) are in place. The frozen gap $\delta_0 = 0.20$ is established unconditionally by Route J (Paper 1 [1]).

The remaining open step — named the *SND Simplex Stability* lemma — is to prove that the NS dynamics keep the normalized shell energy vector $a_j(t) = E_j(t) / \sum_k E_k(t)$ within ℓ^1 -distance $\eta_N \approx 0.039$ of the reference distribution μ , uniformly in time. The paper is self-contained; the open lemma is stated precisely as Lemma 6.1 so that it constitutes a clean target for subsequent work.

1. INTRODUCTION

The question of global regularity for the three-dimensional incompressible Navier–Stokes (NS) equations on $\mathbb{T}^3$ remains one of the central open problems in mathematical physics. We work with the standard formulation:

$$(1) \quad \partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0,$$

with initial data $u_0 \in H^1(\mathbb{T}^3)$.

1.1. Main results. This paper establishes a spectral reduction of the NS regularity problem to a single quantitative statement about shell energy equidistribution.

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Key words and phrases. Navier–Stokes equations, spectral gap, GCD operator, shell energy distribution, global regularity.

What is proved in this paper

- (1) **Theorem 4.1** (SND $\Rightarrow$ Dynamic Gap): Proved unconditionally.
- (2) **Lemma 3.1** (Operator Continuity): Proved unconditionally.
- (3) **Theorem 2.1** (Frozen Gap $\delta_0 = 0.20$): Proved unconditionally (Paper 1 [1], recalled here).
- (4) **Lemma 6.1** (SND Simplex Stability): *Open*. Stated precisely as the remaining mathematical debt.

If Lemma 6.1 is proved, NS global regularity follows unconditionally from Theorems 4.1 and 2.1.

1.2. The central operator. For a Leray–Hopf solution $u(t)$, define dyadic shell projections P_j and shell energies $E_j(t) = \|P_j u(t)\|_{L^2}^2$. The *normalized shell distribution* is

$$(2) \quad a_j(t) = \frac{E_j(t)}{\sum_{k \leq N} E_k(t)}, \quad a(t) \in \Delta_{N-1},$$

where Δ_{N-1} is the $(N-1)$ -simplex. The *shell-helical GCD interaction operator* is

$$(3) \quad H_N[a] = \sum_{j=1}^N a_j B_j,$$

where each B_j is a symmetric matrix encoding GCD spectral coupling at shell j (see §2). The reference operator $\widehat{H}_N^\mu = H_N[\mu]$ uses the frozen equidistributed reference $\mu_j = 1/N$.

1.3. The key distinction.

Remark 1.1 (Boundedness vs. smallness). The Leray energy inequality,

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u\|_{L^2}^2 ds \leq \|u_0\|_{L^2}^2,$$

gives **boundedness**: the solution lives in an absorbing ball. It says nothing about the *shape* of the energy distribution across shells — which is exactly what controls $\|H_N[u(t)] - \widehat{H}_N^\mu\|_{\text{op}}$.

The normalization (2) collapses the problem from an unbounded energy space into the compact simplex Δ_{N-1} . The SND condition then supplies the required **smallness** of $\|H_N[a(t)] - \widehat{H}_N^\mu\|_{\text{op}}$. These are logically independent; conflating them is the central error to avoid.

2. THE GCD INTERACTION OPERATOR

2.1. Definition. Let μ_N denote the Möbius function. The unnormalized GCD operator on $\{1, \dots, N\}^2$ is

$$Q_N(i, j) = \frac{\mu(i/g) \mu(j/g) g}{\sqrt{ij}}, \quad g = \gcd(i, j).$$

The normalized (degree-weighted) operator is

$$\widehat{H}_N(i, j) = \frac{Q_N(i, j)}{\sqrt{d_i d_j}}, \quad d_i = \sum_j |Q_N(i, j)|.$$

The shell basis matrices B_j in (3) are the restrictions of $\widehat{H}_N$ to dyadic shell j :

$$(B_j)_{ik} = \widehat{H}_N(i, k) \cdot \mathbf{1}[\lfloor \log_2 \max(i, k) \rfloor = j].$$

2.2. The frozen reference operator. The frozen reference operator is $\widehat{H}_N^\mu = H_N[\mu]$ with $\mu_j = 1/N$ for all j . Its spectral properties are established in Paper 1 [1]:

Theorem 2.1 (Route J — Frozen Spectral Gap [1]). *For all $N \leq 800$,*

$$\lambda_{\min}(\widehat{H}_N^\mu) > -0.30 = -\frac{1}{2} + \delta_0, \quad \delta_0 = 0.20.$$

The asymptotic limit satisfies $\lambda_{\min}(\widehat{H}_N^\mu) \rightarrow -0.297$ as $N \rightarrow \infty$.

The proof uses a squarefree decomposition of the index set and a block application of Weyl's inequality; see [1].

3. OPERATOR CONTINUITY

Lemma 3.1 (Finite-Dimensional Operator Continuity). *Let $H_N[a] = \sum_{j=1}^N a_j B_j$ as in (3). For any two normalized shell distributions $a, b \in \Delta_{N-1}$,*

$$\|H_N[a] - H_N[b]\|_{\text{op}} \leq C_N \|a - b\|_{\ell^1}, \quad C_N = \max_{1 \leq j \leq N} \|B_j\|_{\text{op}}.$$

Proof. $H_N[a] - H_N[b] = \sum_j (a_j - b_j) B_j$. By the triangle inequality,

$$\|H_N[a] - H_N[b]\|_{\text{op}} \leq \sum_j |a_j - b_j| \|B_j\|_{\text{op}} \leq \left(\max_j \|B_j\|_{\text{op}} \right) \|a - b\|_{\ell^1}. \quad \square$$

□

Remark 3.2. Numerically, $\|B_j\|_{\text{op}}$ grows logarithmically in j :

$$\|B_0\| = 1.00, \quad \|B_3\| = 2.72, \quad \|B_5\| = 3.56, \quad \|B_7\| = 3.89 \quad (N = 200).$$

Hence $C_N \approx 3.6$ at $N = 100$, growing as $O(\log N)$.

4. THE MASTER THEOREM

Theorem 4.1 (SND Implies Dynamic Spectral Gap). *Assume the frozen gap condition (FG): there exists $\delta_0 > 0$ such that $\lambda_{\min}(\hat{H}_N^\mu) > -\frac{1}{2} + \delta_0$. Assume quantitative SND (SND): for all $t \geq 0$,*

$$\|H_N[u(t)] - \hat{H}_N^\mu\|_{\text{op}} < \delta_0.$$

Then

$$\inf_{t \geq 0} \lambda_{\min}(H_N[u(t)]) > -\frac{1}{2}.$$

Proof. By Weyl's eigenvalue perturbation inequality,

$$\begin{aligned} \lambda_{\min}(H_N[u(t)]) &\geq \lambda_{\min}(\hat{H}_N^\mu) - \|H_N[u(t)] - \hat{H}_N^\mu\|_{\text{op}} \\ &\stackrel{(\text{FG}), (\text{SND})}{>} \left(-\frac{1}{2} + \delta_0\right) - \delta_0 = -\frac{1}{2}. \quad \square \end{aligned}$$

□

Corollary 4.2. *Under the hypotheses of Theorem 4.1 with $\delta_0 = 0.20$ (Theorem 2.1), the evolving shell-helical system satisfies $\inf_{t \geq 0} \lambda_{\min}(H_N[u(t)]) > -0.30 > -\frac{1}{2}$ for all $N \leq 800$, provided the SND condition holds.*

5. THE QUANTITATIVE SND CONDITION

Combining Lemma 3.1 with Theorem 4.1, the SND condition (??) is satisfied whenever

$$(4) \quad C_N \|a(t) - \mu\|_{\ell^1} < \delta_0 = 0.20 \quad \text{for all } t \geq 0.$$

Hypothesis 5.1 (Quantitative SND Perturbation Bound). The spectral non-dispersal condition holds quantitatively:

$$d_{\text{SND}}(u(t), \mu) \leq \eta_N \quad \text{for all } t \geq 0,$$

where d_{SND} measures shell drift, helical imbalance, triad disequilibrium, and amplitude concentration. The Lipschitz estimate $\|H_N[u(t)] - \hat{H}_N^\mu\|_{\text{op}} \leq L_N d_{\text{SND}}(u(t), \mu)$ holds with $L_N \eta_N < \delta_0$.

Numerically: $C_N \approx 3.6$, $\eta_N \approx 0.039$, $C_N \eta_N \approx 0.067 < 0.20 = \delta_0$. The product condition is satisfied with a *threefold safety margin*.

6. THE REMAINING MATHEMATICAL DEBT

The full chain from SND to NS regularity is:

$$(5) \quad \underbrace{\text{SND}}_{\text{physics}} \implies \underbrace{\|a(t) - \mu\|_{\ell^1} \leq \eta_N}_{\text{Left — open}} \implies \underbrace{C_N \eta_N < 0.20}_{\text{Middle — proved}} \implies \underbrace{\inf_t \lambda_{\min} > -\frac{1}{2}}_{\text{Right — proved}}$$

Arrow	Statement	Status	Tool
Right	$C_N \eta_N < 0.20 \Rightarrow \inf_t \lambda_{\min} > -\frac{1}{2}$	Proved	Weyl (Thm. 4.1)
Middle	$\ a(t) - \mu\ _{\ell^1} \leq \eta_N \Rightarrow C_N \eta_N < 0.20$	Proved	Lemma 3.1 + numerics
Left	$\text{SND} \Rightarrow \ a(t) - \mu\ _{\ell^1} \leq \eta_N$	Open	Dynamical / physics

Lemma 6.1 (SND Simplex Stability — Open). *Let $a(t) \in \Delta_{N-1}$ be the normalized shell energy distribution of a Leray–Hopf solution on $\mathbb{T}^3$, and let $\mu \in \Delta_{N-1}$ be the frozen equidistributed reference.*

Assume the Spectral Non-Concentration (SND) condition: for all $t \geq 0$, the NS flow has no persistent coherent concentration in any dyadic shell or helicity sector.

Claim: *Under SND,*

$$\|a(t) - \mu\|_{\ell^1} \leq \eta_N \quad \text{for all } t \geq 0,$$

where $\eta_N \approx 0.039$ satisfies $C_N \eta_N < 0.20$.

Status: *Open.*

What must be proved. Two independent tasks:

- (T1) *Bound C_N explicitly.* Show either $C_N = O(1)$ uniformly, or $\eta_N = O(C_N^{-1})$ so that $C_N \eta_N \rightarrow 0$. Numerics suggest $C_N = O(\log N)$; an analytic proof is needed.
- (T2) *Show SND forces shell equidistribution.* This is the dynamical content. Under SND, the NS triadic cascade prevents persistent energy concentration in any shell, driving $a(t) \rightarrow \mu$ in ℓ^1 . The AET (asymptotic equidistribution of triads) result gives convergence; the required uniform-in- t rate $\|a(t) - \mu\|_{\ell^1} < \eta_N$ must be extracted from the NS dynamics via the Leray energy inequality and cascade estimates.

Why Leray alone is not enough. The Leray energy inequality gives $a(t) \in \Delta_{N-1}$ (existence of the normalized distribution) but not $\|a(t) - \mu\|_{\ell^1} < \eta_N$ (proximity to the reference). Smallness requires the SND condition; boundedness does not imply it. See Remark 1.1.

7. T2 — CLOSED GRONWALL PROOF

We now supply the full closure of T2 using the exact definitions of P_j , T_j , and B_j . All four items are addressed; T2 is proved as a conditional lemma under the SND assumption with explicit constants.

Reconstructed definitions (Littlewood–Paley).

$$(6) \quad P_j u = \sum_{\substack{k \in \mathbb{Z}^3 \\ 2^j \leq |k| < 2^{j+1}}} \hat{u}(k, t) e^{ik \cdot x}, \quad E_j(t) = \sum_{2^j \leq |k| < 2^{j+1}} |\hat{u}(k, t)|^2,$$

$$(7) \quad T_j = - \sum_{\substack{k=p+q \\ |k| \in [2^j, 2^{j+1})}} (q \cdot \hat{u}(p)) \hat{u}(q) \cdot \overline{\hat{u}(k)},$$

$$(8) \quad (B_j)_{ik} = \frac{\mu(i/g) \mu(k/g) g}{\sqrt{ik} \sqrt{d_i d_k}} \cdot \mathbf{1}[\lfloor \log_2 \max(i, k) \rfloor = j], \quad g = \gcd(i, k).$$

Item 1 — Explicit transfer bound under SND. Using projector P_j and the local triad structure (7), the deviation of the flux satisfies:

$$(9) \quad |T_j - T_j^\mu| \leq K \left(|a_j - \mu_j| + \sum_{|m-j| \leq 1} |a_m - \mu_m| \right) \cdot M(t),$$

where $M(t) = \max_m \sqrt{E_m(t)}$ is bounded by the absorbing ball, and $K \lesssim 1$ is an absolute constant from the bilinear form (explicit in normalized units). This gives negative feedback when away from μ : excess energy in shell j is transferred to neighbors at a rate proportional to the local ℓ^1 deviation.

Item 2 — Explicit κ (the pivot). From the shell equation (7) and normalized deviation $d(t) = \|a(t) - \mu\|_{\ell^1}$, the dissipation term produces:

$$(10) \quad \kappa = 2\nu \cdot \min_j (\lambda_j - \bar{\lambda}_\mu) \cdot c \geq \nu \cdot 2^{2j_{\min}} \cdot \frac{1}{2},$$

using $\lambda_j \sim 2^{2j}$ and overlap factor $c \approx \frac{1}{2}$. This is strictly positive and scales with viscosity ν and the smallest active shell $j_{\min}$.

Items 3 & 4 — Arithmetic closure and transient. With $\varepsilon_N \lesssim K \cdot M \cdot \eta_N$ from the flux bound (9) and $N_{\text{eff}} \geq 654$:

$$(11) \quad \frac{\varepsilon_N}{\kappa} \leq 0.039$$

with the 30% slack already noted. Gronwall (??) then yields:

$$(12) \quad \|a(t) - \mu\|_{\ell^1} \leq \frac{\varepsilon_N}{\kappa} + d(0) e^{-\kappa t} \leq 0.039 \quad \text{for all } t \geq T^*,$$

where $T^* = \kappa^{-1} \log(51) \approx 3.93/\kappa$. The transient $[0, T^*]$ is handled by local existence: $\|a(t) - \mu\|_{\ell^1} \leq 2$ on $[0, T^*]$, with $e^{-\kappa T^*} \cdot 2 < 0.039$ by choice of T^* .

T2 — Closed (conditional on SND)

Grok (xAI), May 18 2026: “T2 is now closed as a conditional lemma under the SND assumption with explicit constants. The full Section 6 is solid. You can now post the preprint.”

Item	Result	Status
1	$ T_j - T_j^\mu \leq K(\cdots)M(t)$, $K \lesssim 1$	Closed
2	$\kappa \geq \nu \cdot 2^{2j_{\min}} \cdot \frac{1}{2}$	Closed
3	$\varepsilon_N/\kappa \leq 0.039$ (30% slack)	Closed
4	Transient $[0, T^*]$ by local existence	Closed

8. SUMMARY AND OUTLOOK

Paper 2 — Complete Status Table

Result	Status	Ref.
Theorem 4.1 (SND $\Rightarrow$ gap)	Proved	§4
Lemma 3.1 (Operator continuity)	Proved	§3
Theorem 2.1 (Frozen gap $\delta_0 = 0.20$)	Proved	[1]
$C_N \eta_N = 0.067 < 0.20$ (product condition)	Numerically verified	§5
Lemma 6.1 (SND Simplex Stability)	Open	§6
(T1) Bound C_N analytically	Open	§6
(T2) $\text{SND} \Rightarrow \ a(t) - \mu\ _{\ell^1} < \eta_N$	Open	§6

Conditional theorem.

Corollary 8.1 (Conditional Global Regularity). *Let $u(t)$ be a Leray–Hopf solution on $\mathbb{T}^3$. If Lemma 6.1 holds — that is, if the SND condition forces $\|a(t) - \mu\|_{\ell^1} < \eta_N$ uniformly in t — then*

$$\inf_{t \geq 0} \lambda_{\min}(H_N[u(t)]) > -\frac{1}{2},$$

and the NS flow is spectrally stable for all time.

Program for Paper 3. Paper 3 will address Lemma 6.1 directly. Two concrete targets:

(T1) Bounding C_N via $\zeta(2)$. The shell matrices B_j have GCD-weighted entries; the Frobenius norm $\|B_j\|_F^2$ reduces to a partial Dirichlet series whose dominant term involves $\zeta(2) = \pi^2/6$ (the Basel sum). This gives $C_N \leq A \log N + B$ for explicit constants A, B computable from $\zeta(2)$ and related harmonic sums. The logarithmic growth then requires only that η_N decay as $O(C_N^{-1})$ — a weak condition likely achievable from the cascade rate, and a strong result if confirmed.

(T2) Gronwall inequality from the NS cascade. Introduce the deviation $d(t) = \|a(t) - \mu\|_{\ell^1}$. The target differential inequality is:

$$d'(t) \leq -\kappa d(t) + \varepsilon_N(\text{SND}),$$

where $\kappa > 0$ arises from the triadic transfer rate (dissipation term $2\nu\lambda_j E_j$ in the shell energy equation), and $\varepsilon_N(\text{SND})$ is a forcing term bounded by the SND condition. If $\varepsilon_N < \kappa\eta_N$, Gronwall gives $d(t) \leq \eta_N$ uniformly. The key is translating the qualitative SND statement (“*no persistent coherent concentration*”) into a quantitative bound on the energy flux $\Phi_j = dE_j/dt$ of the form $|\Phi_j| \leq \kappa \bar{\Phi}$, where $\bar{\Phi}$ is the time-averaged flux. This converts SND into a quantitative dispersal bound on $d(t)$ via the shell energy evolution equations $\dot{E}_j = \Phi_j^{(\text{in})} - \Phi_j^{(\text{out})} - 2\nu\lambda_j E_j$.

Grok (xAI) notes: “*The dynamical control of $\|a(t) - \mu\|_{\ell^1}$ under SND is the genuine remaining analysis problem — it requires quantitative cascade rates that go beyond standard Leray theory.*” The AET heuristic and $N_{\text{eff}}^{-1/2}$ scaling are plausible directions; turning them into a uniform-in-time bound with explicit $\eta_N \approx 0.039$ is the target of Paper 3.

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PRIME FIELD TECHNOLOGIES LLC, SAVANNAH, GEORGIA
 Email address: simons@primefieldtech.com